

Article

Dative Clitics as Arguments and Adjuncts: A Developmental Perspective on Sentence Processing in Italian Children, Adolescents, and Adults

Matteo Greco ^{1,*} , Veronica D'Alesio ^{1,2}  and Anna Teresa Porrini ¹ 

¹ Scuola Universitaria Superiore IUSS Pavia, 27100 Pavia, Italy; veronica.dalesio@iusspavia.it (V.D.); anna.porrini@iusspavia.it (A.T.P.)

² Department of Letters and Modern Cultures, Università degli Studi di Roma “La Sapienza”, 00185 Roma, Italy; veronica.dalesio@uniroma1.it

* Correspondence: matteo.greco@iusspavia.it

Abstract

This study investigates the processing of Italian verb constructions containing preverbal dative clitics that function either as arguments or as adjuncts, adopting a developmental perspective across Italian-speaking children, adolescents, and adults. In a self-paced reading experiment, adults and fourteen-year-old adolescents read trivalent verbs selecting a dative argument ('Martina *mi* ha prestato una bicicletta.'; Eng.: 'Martina lent *me* a bicycle.') faster than bivalent verbs licensing a benefactive dative ('Martina *mi* ha riparato una bicicletta.'; Eng.: 'Martina repaired *me* a bicycle), in line with the Argument Structure Hypothesis, which predicts lower processing costs for lexically selected arguments than for syntactically computed adjuncts. In contrast, eleven-year-old children showed no processing differences between the two constructions. A second experiment using eye-tracking with ten-year-old children replicated this null effect, indicating that the absence of sensitivity to the argument–adjunct distinction at this age is not due to methodological factors. Instead, the findings suggest a developmental trajectory, with adult-like processing emerging only in early adolescence. Children's performance appears to be influenced by grammatical competence rather than by verb argument structure per se. Overall, the results highlight the role of grammatical maturation in real-time syntactic processing.

Keywords: dative clitics; argument–adjunct distinction; sentence processing; language development; self-paced reading; eye-tracking



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1. Introduction

Understanding how syntax develops during childhood requires not only examining what structures children know, as seen through their correct use and comprehension, but also how such structures are processed in real time. While most developmental research has focused on the former aspect, investigating child syntax mainly in the earliest years of life, there are aspects of syntactic development, as well as the neural circuits that support it, that continue to mature well into adolescence (Nippold, 1998; Schneider & Maguire, 2019). Investigating real-time sentence processing at later ages could therefore prove useful in tracing the developmental trajectory of syntactic competence. This is particularly important when processing happens during reading, given how most educational practices include reading as a core, fundamental skill. By linking formal linguistic theory on syntax with psycholinguistic evidence, one of the aims of the present work is to inform educational

practices, providing insights into how syntactic structures are processed during reading, with implications for supporting developing readers. More specifically, the present study focuses on how readers process different types of syntactic dependencies, with particular attention to the distinction between arguments and adjuncts. By examining how these two classes of constituents are integrated during real-time sentence comprehension, we aim to shed light on their respective roles in the development of syntactic processing during reading.

1.1. The Argument vs. Adjunct Distinction

Dative DPs are traditionally divided into two major classes across languages (Hale & Keyser, 2002). On the one hand, there are *core/argumental* datives, which are integrated into the predicate's thematic grid. They function as arguments, as in ditransitive constructions with verbs such as *donate*. On the other hand, languages also exhibit *non-core/non-argumental* datives, elements that do not arise from the verb's argument structure, adding optional information, as in transitive constructions with verbs such as *melt*:

- (1) a. Bill donates some money *to John*.
b. Bill melted some ice *for John*.

In sentence (1a), the phrase *John* represents one of the arguments of the verb *to donate*, the goal of the action, whereas, in (1b), it does not, falling into the class of “adjuncts”. Adjuncts can be freely added to a clause, bringing some extra information to the sentences in which they occur (Marantz, 1984, 1993): in (1b), the prepositional phrase points out the individual, *John*, who is the beneficiary of the action described by the verb (Folli & Harley, 2006). This is the reason why cases like these are called “benefactives”, which display particularly complex and significant properties (Jaeggli, 1982; Perlmutter, 1971; Rivas, 1977; Greco, 2025).

Languages also display different strategies to add both goal arguments and benefactive adjuncts, such as by the use of pronouns. Consider, again, the sentence in (1): the phrase *to/for John* can be substituted by the use of the pronoun *him*.

- (2) a. Bill donates *him* some money.
b. Bill melted *him* some ice.

Many works, starting with Hale and Keyser's (1993) seminal book, have investigated the syntactic operations and relations between (1) and (2). A longstanding question in the literature concerns how goal arguments and benefactive datives are licensed, particularly in the latter case when no verb semantically selects them, nor assigns a theta-role. A widely explored line of analysis proposes that both datives are introduced into the clause by an independent functional projection—an Applicative Phrase—whose head is responsible for selecting and licensing the additional DP (Cuervo, 2003; Marantz, 1984, 1993; Pylkkänen, 2002, 2008). According to this view, two different types of applicative phrases introduce argument and adjunct items, i.e., high and low Applicative Phrases (ApplPs)—a view also anticipated in Baker (1996). Benefactives fall under the ‘high applicatives’ class, since they encode a relation between an event and an individual (such as the melting of ice and a third person in the sentence in 2b). On the other hand, goal arguments fall under the ‘low applicatives’ class, since they encode a relation between two individuals/elements within the same event: one introduced by the applicative head and the other serving as the verb's direct object, as is typical in ditransitive patterns (2a). Functionally, a high applicative contributes an additional (optional) participant directly to the event described by the verb, effectively introducing an extra external argument. Low applicatives, by contrast, are required by the verbal lexical entry, instantiating a transfer-of-possession relation with the direct object: the money in (2a) becomes transferred to the entity expressed by the

dative pronoun *him*. This high–low split is further supported by syntactic and semantic phenomena (see the cited works for a detailed description).

Many studies point out differences in the processing of arguments and adjuncts (Tutunjian & Boland, 2008), since they rely on different semantic and syntactic strategies, in line with the high–low applicative approach, with adjuncts requiring greater effort to be integrated into the sentence. For example, in Boland and Blodgett’s (2006) eye-movement study, readers show shorter initial processing times for prepositional phrases functioning as arguments than for those serving as adjuncts, suggesting that argument structures are easier to process. This leads some theories, such as the Argument Structure Hypothesis (ASH), to posit a distinction between arguments—required by the lexical representation of the verb—and adjuncts—incorporated by means of general, non-lexical syntactic rules such as phrase-structure configurations (Grimshaw, 1990; Boland & Boehm-Jernigan, 1998), since they are not required by any lexical head. Constructions that include either arguments or adjuncts are, therefore, not only structurally different, but also different in their processing, since they are assumed to be, respectively, lexically accessed or syntactically computed.

Even though such approaches are not without issues—for example, the distinction between the lexicon and syntactic computation can be fuzzy (see Borer (2005) for a detailed discussion)—psycholinguistic data still show differences in their processing. Further evidence comes from Lee and Thompson’s (2011) eye-tracking study with both healthy and agrammatic aphasic speakers, showing that participants’ eye movements during a production task involving sentences containing a goal argument (e.g., “*The mother is applying lotion to the baby*”) and a beneficiary adjunct (e.g., “*The mother is choosing lotion for the baby*”) reveal longer gaze durations and more gaze shifts for the latter than for the former.

The argument vs. adjunct elaboration has been further investigated using both eye-tracking (see, among many others, Kennison, 2002; Boland, 2005; Traxler, 2008, and references therein) and self-paced reading paradigms (Clifton et al., 1991; Schütze & Gibson, 1999), which consistently show longer fixation and reading times for adjuncts than for arguments. A similar contrast also appears in recalling tasks, where core arguments—such as direct objects—are recalled more accurately than both temporal and locative adjuncts (Chromý & Vojvodić, 2024). Adjuncts can therefore be considered more demanding to process than arguments.

Crucially, languages can employ other strategies to introduce goal or benefactive participants into a sentence, such as the use of clitics. This paper will address this specific phenomenon.

1.2. Dative Clitics as Arguments or Adjuncts

Languages can employ a number of strategies to introduce goal or benefactive participants, some of which have been less explored in the experimental literature. One such strategy is the use of dative clitics, as in the following Italian examples:

- (3) a. Martina | *mi* ha | prestato | una bicicletta | bellissima. (Argument)
 Martina | to.meDATIVE.CLITIC has | lent | a bicycle | wonderful.
 ‘Martina lent me a wonderful bicycle.’
 b. Martina | *mi* ha | riparato | una bicicletta | bellissima. (Adjunct)
 Martina | for.meDATIVE.CLITIC has | repaired | a bicycle | wonderful.
 ‘Martina repaired a wonderful bicycle for me.’

In (3a), *mi* expresses the goal argument of the trivalent verb *prestare* ‘lend’, whereas, in (3b), it expresses the beneficiary of the repairing event¹. Besides the argument/adjunct distinction, another grammatical operation is involved here, namely, *cliticisation* (Kayne, 1975; Cinque, 1990; Benincà & Cinque, 2024). In fact, in (3), the clitics precede the finite verb, whereas their default position is post-verbal, as witnessed by the corresponding prepositional phrase *a me* (‘to me’) in (4), suggesting that a movement operation has occurred:

- (4) a. Martina | ha | prestato | una bicicletta | bellissima | a me. (Argument)
 Martina | has | lent | a bicycle | wonderful | to me
 ‘Martina lent a wonderful bicycle to me.’
 b. Martina | ha | riparato | una bicicletta | bellissima | a me. (Adjunct)
 Martina | has | repaired | a bicycle | wonderful | to me
 ‘Martina repaired a wonderful bicycle for me.’

The use of clitics has been extensively investigated in both theoretical and experimental literature, showing that they are already acquired starting from the early stages (McKee, 1992; Schaeffer, 2000; Tedeschi, 2009; Varlokosta et al., 2016), throughout childhood, and up to the later stages, i.e., 10-year-old (see Moscati et al., 2023 and the references therein), particularly for some challenging cases like reconstruction (Cardinaletti et al., 2024).

A similar acquisition pattern is found in adults in L2 cases (Santoro, 2007). Interestingly, some differences within the same clitics system can arise, such as in primary-school-age children, due to both different variables and phi-features clitics show (Caprin & Guasti, 2009; Dispaldro et al., 2014; Delage et al., 2016): case, such as accusative vs. dative, gender vs. gender-free, first/second-person vs. third, etc. For example, third-person singular dative clitic pronouns are more produced than third-person accusative forms when reconstruction is required in Italian (Cardinaletti et al., 2021).

Besides these cases, the difference between goal argumental and benefactive non-argumental dative clitics remains largely undiscovered, with some exceptions, such as in L2 and heritage speakers in syntactic priming contexts (Hurtado & Montrul, 2021). More specifically, previous research shows that children can associate verbs with their argument structure and select semantically appropriate thematic roles based on verb meaning from a relatively young age (Peter et al., 2015); less is known about how syntactically specified their representations of verb structure are, and how the distinction between verb arguments and adjuncts is acquired over the course of development. Therefore, the Italian clitic system represents an ideal testing phenomenon because a single surface form, e.g., *mi* (‘to me’) in (3), can map onto two fundamentally distinct syntactic configurations. On one hand, a clitic can serve as a full argument lexically required by a trivalent verb’s thematic grid (e.g., *prestare* / ‘to lend’). On the other hand, it can function as an optional benefactive adjunct licensed by a bivalent verb (e.g., *riparare* / ‘to repair’), without requiring any kind of morphological alteration between the two uses (such as the alternation between *to* and *for* in Lee and Thompson’s (2011) study; e.g., ‘The mother is choosing lotion *to/for* the baby’). This makes clitics a new and interesting phenomenon to be used in order to check the different processing of argument vs. adjunct elements.

The aim of this article is therefore to examine the developmental trajectory from childhood to adulthood in order to understand how these differences in syntactic maturation unfold.

2. The Present Study

Given the theoretical and experimental literature presented on the distinction between argument and adjunct structures, as well as what is known about dative clitics and their acquisition, two main questions are addressed in the present study:

- i. Are verb constructions processed differently depending on the argument/adjunct status of their structural elements—showing longer fixation and reading times for adjuncts as compared to arguments—even when the relevant elements are realised as clitics, rather than as prepositional phrases? This is particularly interesting since, through the use of clitics, arguments and adjuncts can be realised through the exact same surface form (e.g., *mi*, ‘to me’)².
- ii. Are older children and adolescents adult-like in parsing such constructions?

To address these questions, a first experiment using self-paced reading (SPR) was designed. This online methodology enabled us to test adults, adolescents, and children's processing of sentences presenting argument (3a) and adjunct (3b) structures while reading. Within this paradigm, participants read the sentences in a segment-by-segment (or chunk-by-chunk) format. Each chunk appeared only after the participant pressed a button, at which point the preceding one disappeared. This procedure ensured that readers would set their own reading speed, rendering comprehension self-paced. The latency between the presentation of each chunk and the button press (i.e., reaction time, RT) is used as a measure of processing effort (Jegerski, 2014; Keating & Jegerski, 2015).

Concerning question (i), if argument and adjunct structures are processed differently even when the relevant elements are realised as clitics, then we would hypothesise longer RTs for adjunct structures. Specifically, a processing cost should be seen once participants reach the verb, which can be either a trivalent verb selecting an argument (3a) or a bivalent verb allowing the presence of a benefactive adjunct (3b). As for question (ii), if older children and/or adolescents are indeed competent in the processing of these constructions, given their knowledge of clitics and of the argument structure, we anticipate that they will show adult-like effects, exhibiting similar RT increases on the verb segment in adjunct structures. Any divergence from this pattern, however, could suggest that the efficient processing of clitic-based argument structure distinctions continues to develop during late childhood and potentially adolescence, opening new developmental research avenues.

The experimental stimuli for both experiments, as well as the datasets containing the results and the full analysis scripts, can be found in our online repository on OSF (<https://osf.io/8mvua>, accessed on 24 June 2026).

3. Experiment 1: Materials and Methods

3.1. Participants

Three groups of native Italian speakers participated in the experiment.

1. The first group consisted of 85 adults (45 female, 40 male; mean age = 34, minimum = 25, maximum = 49, SD = 7), recruited online through Prolific.com and paid approximately 2€ for their participation. All participants reported normal or corrected-to-normal vision and no known language or learning impairments.
2. The second participant group was initially composed of 98 14-year-old adolescents, recruited and tested at their high school in *Scuola Secondaria IIS Falcone-Righi* of Corsico (Italy). Of these, sixteen were later excluded due to having a diagnosis of dyslexia, and another two were excluded due to low accuracy (<70%) in their responses to comprehension questions after filler items, under the assumption that they were not paying sufficient attention to the task. The final group was composed of 80 adolescents (21 female, 1 who preferred not to say, 58 male; mean age = 14;5, minimum = 13;10, maximum = 15;5, SD = 0;4).
3. The third group consisted of 55 11-year-old children, recruited and tested at their middle school in *Scuola primaria San Giorgio* of Pavia (Italy). Of these, four were later excluded due to having a diagnosis of dyslexia and another three were excluded due to low accuracy (<70%) in their responses to comprehension questions after filler items, under the assumption that they were not paying sufficient attention to the task. The final group was composed of 48 children (24 female, 24 male; mean age = 11;4, minimum = 10;10, maximum = 11;10, SD = 0;3).

The study was approved by the Ethics Committee of the University of Pavia/Dipartimento di Scienze del Sistema Nervoso e del Comportamento, and informed consent was obtained from adult participants and the parents or legal guardians of the minors; assent was also

obtained from adolescent and child participants, and it was made clear to them that they could withdraw from participating in the study at any moment, without consequence.

3.2. Materials

The experimental stimuli consisted of ten pairs of Italian sentences constructed to contrast argumental vs. adjunct uses of the same preverbal dative clitic *mi* (“to/for me”). In each pair, the sentences were identical except for the verb presented in each condition. In the argument condition, the sentence contained a trivalent verb that selects the preverbal dative clitic *mi* as its argument. In the adjunct condition, the sentence contained a bivalent verb that could take a benefactive adjunct in the form of the preverbal dative clitic *mi*. This allowed the surface form of the clitic to remain identical across conditions, while the underlying syntactic status of the clitic varied. Since trivalent verbs belong to particular semantic categories, such as verbs of saying and giving (Tesnière, 1959, pp. 255–258), sentences were built starting from the argument condition. Afterwards, 10 bivalent verbs were selected considering the overall semantic fitness of the sentence and the length of their past participle form, so that verbs belonging to each critical duplet had a maximal difference of one character.

An example of each condition is provided below (each sentence is divided into the chunks used in the experimental display, as will be specified later):

Argument condition (trivalent verb):

Questo pomeriggio | al parco | Martina | mi ha | prestato | una bicicletta | bellissima.
This afternoon | at the park | Martina | to.meDATIVE has | lent | a bicycle | wonderful.
‘This morning at the park, Martina lent a wonderful bicycle to me.’

Adjunct condition (bivalent verb):

Questo pomeriggio | al parco | Martina | mi ha | riparato | una bicicletta | bellissima.
This afternoon | at the park | Martina | to.meDATIVE has | repaired | a bicycle | wonderful.
‘This morning at the park, Martina repaired a wonderful bicycle for me.’

The ten experimental pairs were divided across two lists to ensure that each participant saw each sentence in only one of the two conditions, and they were spaced out within the task by 70 filler items. They were of comparable length to the critical items and a good portion of them contained both male and female personal names, so that critical items could blend in with respect to this possibly salient feature as well.

3.3. Design

The experiment was implemented in a masked, chunk-by-chunk, non-cumulative self-paced reading (SPR) paradigm. Each sentence was divided into seven chunks, roughly corresponding to major syntactic constituents. The first two chunks of each sentence presented information about the location and/or timing of the event described; they were meant to make the sentence more natural. The following chunks always presented the same structure (example chunks are provided from sentences (3a) and (3b) above):

Chunk 3. A proper name serving as subject (e.g., Martina);

Chunk 4. The dative clitic and the auxiliary of the main verb (e.g., mi ha; Eng. to/for me has);

Chunk 5. The past participle of the main verb (*critical chunk*; e.g., prestato/riparato; Eng. lent/repaired);

Chunk 6. An indefinite determiner and noun, serving as direct object (e.g., una bicicletta; Eng. a bicycle);

Chunk 7. An object modifier (e.g., Bellissima; Eng. wonderful).

Chunks were initially hidden under a string of dashes. Participants revealed each chunk by pressing the spacebar, at which point the previous chunk was re-masked. Reading

times were recorded for each chunk, and all items (both critical and fillers) were followed by yes/no comprehension questions. The questions were not meant to target the participants' understanding of the dative clitic, but simply to ensure that participants were paying attention to the stimuli and reading for comprehension. To this aim, questions never targeted the predicate (i.e., the critical region), but rather asked for information provided either at the beginning (e.g., *'Dopo l'esame della patente Gabriele mi ha cucinato un pranzo memorabile.'*, Q: *'L'esame era per la patente?'*; 'After the driving licence exam Gabriele cooked for me an unforgettable lunch.' Q: 'Was the exam for the driving licence?') or at the end (e.g., *'Prima dell'esame Margherita mi ha inviato un messaggio veramente complicato.'* Q: *'Il messaggio di Margherita era semplice?'*; 'Before the exam Margherita sent me a really complicated message.', Q: 'Was the message from Margherita simple?') of the sentence,—that is, the chunks that were not considered for analysis.

3.4. Procedure

The experiment was carried out online on PennController for Internet Based Experiments (Zehr & Schwarz, 2018). Adults accessed it through the Prolific platform and were tested individually and remotely. Adolescents and children were tested collectively in school computer rooms, after accessing the experiment through a link, with each participant completing the task independently on their monitor. The same presentation of the stimuli was retained across all sessions—that is, all items were displayed on a single line of text written in a monospaced font (Courier New). Before the experimental session, participants completed a short practice block to familiarise themselves with the task and interface. During this practice phase, participants were given written feedback on whether they answered the comprehension questions correctly. However, no feedback was given during the experimental phase. Participants were instructed to read at a natural pace and respond as accurately as possible to comprehension questions. The experimental session was divided into two blocks, between which participants could take a break if they wished to, and items were presented in random order. The task lasted approximately 15 min for adults and 20–30 min for children and adolescents.

3.5. Results

3.5.1. Data Pre-Processing

As a first step, to exclude items that were not read attentively, items with incorrect responses to comprehension questions were excluded from any analysis. Reaction times (RTs) then underwent two rounds of data cleaning, each targeting different sources of variability. First, extreme values that did not plausibly reflect processing effort were excluded. This included any RT inferior to 100 ms, while age-specific thresholds were applied to remove slow responses, reflecting expected developmental differences in baseline reading speed. These thresholds were established by inspecting the average reading times for each chunk in each age group and setting an upper limit corresponding approximately to three times the mean value. This procedure yielded proportionally higher limits the younger the participants were. Second, to reduce the influence of residual participant-level outliers that could disproportionately affect statistical estimates, chunk RTs more than two standard deviations above the mean for each participant in each condition were removed, following standard RT trimming procedures commonly used in reaction time research (Ratcliff, 1993). As a final step of pre-processing, RTs were log-transformed prior to analysis to approximate a normal distribution, in line with the approach to linguistic RT data proposed by Baayen (2008).

3.5.2. Data Analysis

The data analysis focused on the critical chunk, corresponding to the past participle, and the chunk following it (Chunks 5 and 6). A linear mixed model was fitted for each of the chunks under consideration: the log-transformed RTs were analysed as a dependent variable, while the condition (argument vs. adjunct) was the main fixed factor. The argument condition was coded as 1, while the adjunct condition was coded as -1 . For the critical chunk (Chunk 5), the frequency of the verb used was also considered as a fixed factor (while verb length was controlled for across items). Verb frequency was extracted from the Italian corpus Paisà (Lyding et al., 2014) through the corresponding freely accessible online platform (<https://www.corpusitaliano.it/>, accessed on 24 June 2026). The random effects structure was determined in a data-driven manner for each linear model, starting with the maximal random effects model (following Barr et al., 2013), which included intercepts and slopes for participants and items. In case of singularity fit issues, the random structure was then simplified by systematically removing random effects one at a time, guided by the results of a randomised principal component analysis until the issue was resolved.

3.5.3. Results

The mean RTs for Chunks 4 to 7 in the two experimental conditions can be seen in Figure 1, divided by age group. The full outputs of the models can be found in the Supplementary Materials.

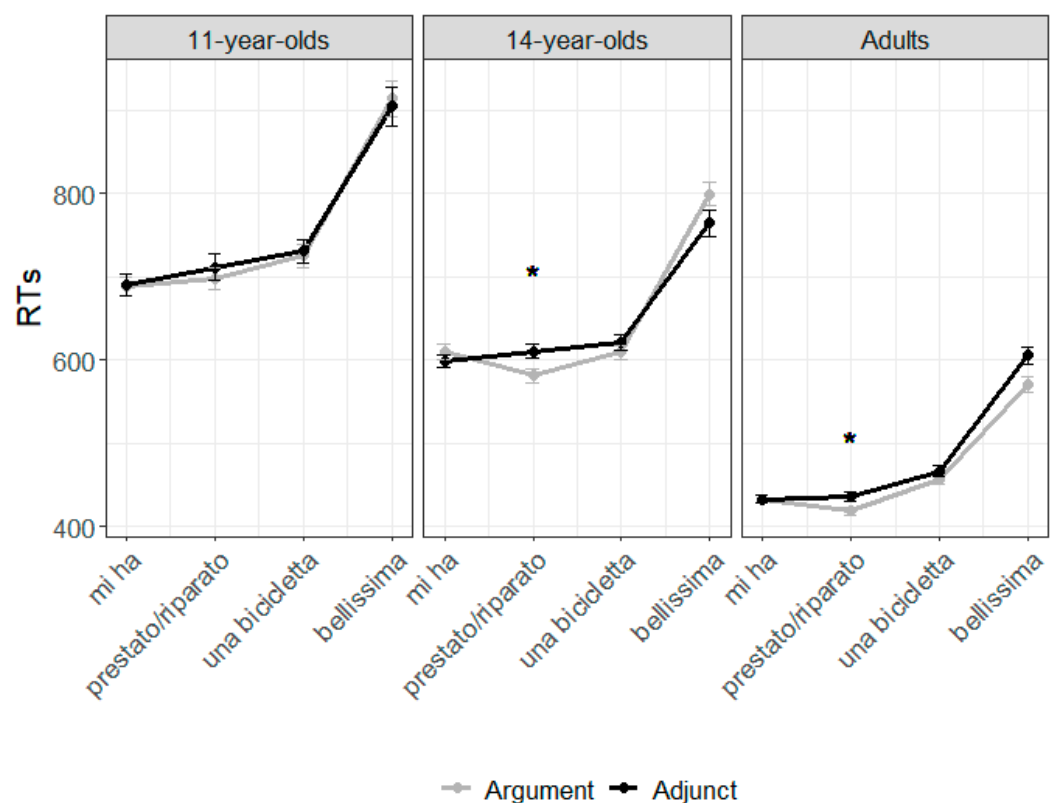


Figure 1. Mean RTs in the two experimental conditions for each age group, with error bars established based on standard errors. Asterisks are drawn based on the statistical analyses reported to signal significant ($p < 0.05$ *) differences between the two experimental conditions.

For the critical chunk (Chunk 5), presenting the past participle of the trivalent (*prestato*, 'lent') or bivalent (*riparato*, 'fixed') verb, the results show a significant effect for condition in adults ($\beta = -0.01$, $SE = 0.01$; $t = -2.32$, $p = 0.023$) and adolescents ($\beta = -0.02$, $SE = 0.01$; $t = -2.29$, $p = 0.023$): trivalent verbs that included dative clitics in their argumental structure

were read faster after dative clitics compared to bivalent ones. Conversely, no effect of condition can be seen for children ($p = 0.962$). Verb frequency did not affect the RTs in any of the three groups. No effect of condition is observed on Chunk 6 in any of the three groups (for adults: $p = 0.511$; for adolescents: $p = 0.434$; for children: $p = 0.915$).

3.6. Interim Discussion

In adults and adolescents, we found longer RTs on bivalent verbs that follow a benefactive dative compared to trivalent verbs that take the dative clitic as their argument, and, in addition, a marginal effect in the same direction is seen for adults. Our findings support the distinction between arguments and adjuncts as well as our initial hypothesis that verb constructions are processed differently based on the argument/adjunct status of their structure elements, even when arguments/adjuncts are realised as clitics (research question (i)). The additional effort detected for the processing of bivalent verbs supports the prediction that arguments and adjuncts differ in their processing demands: while arguments can be retrieved from the verb's lexical entry without requiring additional syntactic computation, adjuncts must be integrated, as they are not encoded in the lexical entry. This additional syntactic burden specific to adjunct integration results in increased processing effort, consistent with the ASH (Boland & Blodgett, 2006). Crucially, this does not imply that argument processing is purely lexical; rather, it reflects the extra syntactic computation uniquely required for adjunct integration.

Importantly, however, while adolescents appear to be (near-)adult-like in parsing such constructions, 11-year-old children did not show any detectable differences in their processing of trivalent and bivalent verbs following a dative clitic (research question (ii)).

Before drawing our conclusion from the data, we decided to further investigate this developmental pattern, especially considering that, according to Lee and Thompson (2011), processing differences between arguments and adjuncts sometimes do not show up in behavioural data, but emerge when looking at eye movements. Given that our youngest group generally exhibited slower and more variable reading times, which might have reflected less automatised decoding skills, the forced linear progression of SPR may potentially have obscured genuine processing differences to a greater degree for them than it did for the other two groups. We therefore proceeded by conducting a second experiment in which we tested a new group of 10-year-old children using eye-tracking during reading. As highlighted by Paolazzi et al. (2022), the reading behaviour elicited in SPR differs from natural reading conditions: participants must advance linearly through the sentence and cannot re-read previously presented segments, even when comprehension difficulties arise. Although SPR is well-suited for identifying early processing effects, it may provide an incomplete picture of later-stage processes involved in integration or reanalysis. In contrast, eye-tracking allows for the continuous measurement of eye movements during reading, providing a more ecologically valid and fine-grained window into processing dynamics (Rayner, 1998; Clifton et al., 2007). By using eye-tracking, we aimed to assess whether the absence of condition effects in children's SPR data reflected a genuine lack of sensitivity to the distinction between argument and adjunct constructions. Furthermore, a measure of children's grammar and vocabulary, as well as their working memory capacity, was collected to try and better understand children's processing patterns.

4. Experiment 2: Methods and Materials

4.1. Participants

One group of 10-year-old children participated in the experiment. The group was composed of 35 10-year-old children, recruited and tested at their elementary school *San Giorgio* in Pavia (Italy) (11 female, 24 male; mean age = 10;5, minimum = 9;9, maximum

= 10;10, SD = 0.3). None of the children had to be excluded due to low accuracy on the filler items. The study was approved by the Ethics Committee of the University of Pavia—*Dipartimento di Scienze del Sistema Nervoso e del Comportamento*, and informed consent was obtained from the parents or legal guardians of the minors; assent was also obtained from the children, and it was made clear to them that they could withdraw from participating in the study at any moment, without consequence. To ensure comparability with the 11-year-old group tested in Experiment 1, both groups were assessed on two standardised measures from the BVL4-12 battery for language assessment (Marini et al., 2015): receptive vocabulary and morphosyntactic comprehension. As shown in Table 1, the two groups did not differ significantly on either measure (vocabulary: $p = 0.602$; morphosyntax: $p = 0.080$), indicating comparable linguistic abilities.

Table 1. Performance on the BVL4-12 for 11-year-old children (Experiment 1) and 10-year-old children (Experiment 2). For each test, mean raw scores and SD are reported on the first row and mean Z scores on the second. The third column reports the p -values from a Mann–Whitney U test comparing Z scores between the two groups.

	11-Year-Olds (Experiment 1)	10-Year-Olds (Experiment 2)	Mann–Whitney U Test
Receptive vocabulary (tot. 42)	37.4 (1.2) $z = -0.44$	36.6 (4.1) $z = -0.17$	$p = 0.602$
Morphosyntactic comprehension (tot. 40)	37.5 (2.3) $z = -0.38$	37.9 (1.7) $z = -0.02$	$p = 0.080$

4.2. Materials

The experimental stimuli were the same as Experiment 1, except for a minor modification meant to adjust the stimuli to the new methodology. Namely, the first two chunks of each item were omitted to ensure that the stimuli were short and presentable in a single line on the screen in a 32-point font, to facilitate reading for children. The 20 experimental items were spaced out throughout the task by 70 filler items, and an additional 4 filler items were used in the practice block.

4.3. Design

The experiment was built in Experiment Builder and conducted using an SR Research Eyelink 1000. Reading was binocular, but data was collected from the dominant eye at a sampling rate of 1000 Hz. Sentences were displayed in 32-point monospaced Courier New font, with black text on a light grey background. Participants could take as much time as they wished to read the sentence. They were instructed to press the spacebar once they were done reading to proceed to the next sentence. Because the eye-tracking paradigm can be more demanding for participants compared to self-paced reading, in Experiment 2, comprehension questions meant to ensure reading for comprehension were included after only a quarter of the items, all of which were filler sentences.

4.4. Procedure

Prior to the eye-tracking task, the BVL4-12 (Marini et al., 2015) was administered to assess the participants' receptive vocabulary and grammar; additionally, a Digit Span test (forward and backward) was also administered as a measure of working memory (WM). Then, for the eye-tracking task, participants were seated approximately 60 cm from the screen, with a chin-rest to minimise head movements. Calibration was performed on a 9-point grid before the practice trials and repeated as necessary. The experiment

started with a practice block followed by two experimental blocks, during which trials were presented in random order across the two blocks using the corresponding feature in Experiment Builder. Children were encouraged to take a break between the blocks in case they were tired of the chin-rest. Calibration took place again after the break if they moved. Each trial began with a fixation point for drift correction, aligned with the start of the upcoming sentence on the left and centred vertically. The task lasted approximately 30 min.

4.5. Results

4.5.1. Data Pre-Processing

Before data extraction, fixation data were pre-processed within the SR Research's DataViewer software (version 4.3) following their four-step cleaning procedure. In the final stage of this process, after merging short fixations into neighbouring ones, remaining fixations shorter than 80 and fixations longer than 1400 milliseconds were removed. Later, before data analysis, Reading Time measures (RTs) exceeding 2 standard deviations from each participant's mean in each condition were excluded. As a final step of pre-processing, RTs were log-transformed following the same reasoning applied for Experiment 1.

4.5.2. Data Analysis

Three Areas of Interest (AoI), automatically segmented at the word-level by Experiment Builder, were taken into consideration for data analysis:

AoI 1. The dative clitic (*mi*);

AoI 2. The past participle of the bivalent or trivalent verb (*critical region*);

AoI 3. The last word of each sentence.

While the three AoIs were all considered for analysis, we refer to AoI 2 as the critical region, consistent with our initial experimental hypothesis and previous experiment (see Section 2). Since the sentence condition (i.e., argument/adjunct) is only revealed once the verb is reached, a difference in reading times was hypothesised in this area, an outcome that was indeed confirmed by our results in the adult and the adolescent participants alike.

For AoI 1, as it preceded the critical region during reading, we extracted only late measures: Total reading time (the sum of the durations of fixations made on an AoI) as a measure of sentence processing difficulty, and Regression in count (the number of regressive fixation on an AoI) to observe possible revisiting of this area after the critical region had been read. Additionally, for AoI 2, we extracted the same late measures with the addition of Regression path duration (the sum of the durations of all fixations made from the entry to an AoI until its exit from the right, including any regressive fixations to its left), as well as early processing measures: First fixation duration (the duration of the first fixation made on an AoI) and Gaze duration (the sum of the durations of all fixations made on an AoI during first pass reading). For AoI 3, we focused only on Regression out count (the number of regressive launches from an AoI). Regression out count was considered as an indicator of the effort required to process the sentence. Given that the final words in each item differed significantly from each other, it was deemed inappropriate to also collect durational measures (i.e., Total reading time, and Regression path duration) to assess wrap-up effects on this AoI.

Generalised linear mixed models (GLMMs) with a Poisson distribution were used for discrete count variables (Regression in and out), while, for RTs, which included First fixation duration, Gaze duration, Total reading time, and Regression path duration, linear mixed-effects models (LMMs) were fitted. The fixed factors for each model were the experimental condition (argument vs. adjunct), the standardised grammar, vocabulary, and WM scores, and the interactions between each of these individual measures and the

experimental condition. As in Experiment 1, the frequency of the verb was added for AoI 2. The random structure was established as in Experiment 1.

4.5.3. Results

The full outputs of the models can be found in the Supplementary Materials.

AoI 1. No significant effect of condition on the dative clitic *mi* was detected in any of the measures taken into consideration (for Total reading time: $p = 0.344$; for Regression in count: $p = 0.417$). However, grammar scores did have an effect on Regression in count ($\beta = -0.28$, $SE = 0.13$; $z = -2.14$, $p = 0.032$): the lower the child's grammatical competence, the more regressions in on the dative clitic, as can be appreciated by looking at Figure 2. This is the only fixed factor that was shown to have an effect in both models.

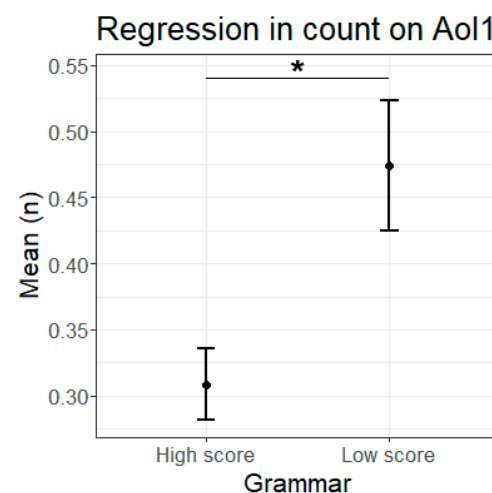


Figure 2. Mean number of Regressions into AoI 1 (the dative clitic) based on standardised grammatical competence, where a low score corresponds to a score below 0 and a high score to one equal to or above 0. Asterisks are drawn based on the statistical analyses reported to signal significant ($p < 0.05$ *) differences between the two grammar levels.

AoI 2. On the critical region (the past participle of the trivalent/bivalent verb), no significant effect of condition is seen in any of the measures taken (for First fixation duration: $p = 0.283$; for Gaze duration: $p = 0.968$; for Regression path duration: $p = 0.158$; for Total reading time: $p = 0.128$). A representation of the results for early (Gaze duration) and late (Total reading time) measures can be found in Figure 3. The only marginally significant effect was detected on Regression in count ($\beta = 0.17$, $SE = 0.13$; $z = 1.94$, $p = 0.052$). Aside from being only marginally significant, this effect was in the opposite direction compared to the one expected and to what was seen for other measures, as it signalled more regressions in on the verb in the argument structure. Verb frequency was the only fixed factor significantly affected reading measures, specifically Gaze duration ($\beta = -0.08$, $SE = 0.03$; $t = -2.76$, $p = 0.009$) and Regression path duration ($\beta = -0.13$, $SE = 0.03$; $t = -3.84$, $p < 0.001$), with shorter RTs for more frequent verbs.

AoI 3. No significant effect of condition, or of other fixed factors, was seen in terms of the number of Regressions out from the last word of the sentence ($p = 0.688$).

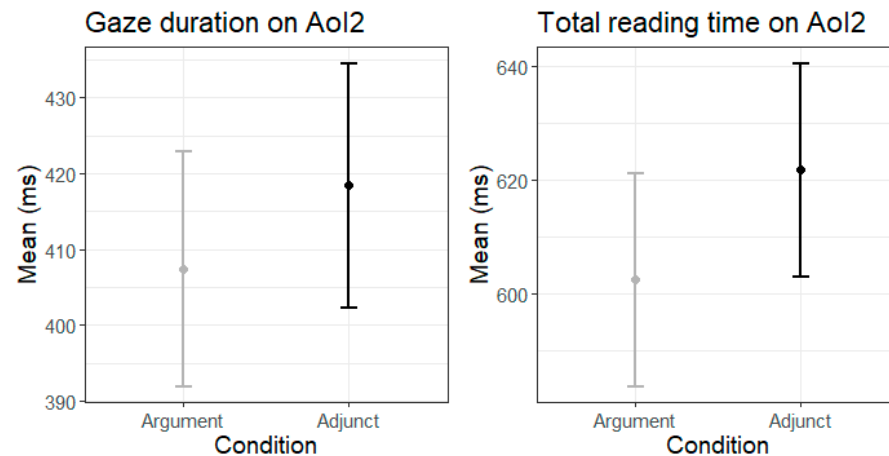


Figure 3. Mean RTs in the two experimental conditions for Gaze duration and Total reading time on AoI 2 (the critical region, corresponding to the past participle of the bivalent/trivalent verb). Asterisks and dots are absent as the statistical analyses did not report significant differences between the two experimental conditions.

4.6. Interim Discussion

The results of Experiment 2 show no significant differences in children’s processing of trivalent and bivalent verbs following a dative clitic in any of the reading measures examined. Across all the AoIs under examination, 10-year-old children showed comparable Regression counts and RTs in the two experimental conditions. The only significant effects observed concerned verb frequency, with shorter fixation durations and regression path durations for more frequent verbs, and grammar score, as children with higher scores performed fewer regressions in on the clitic, regardless of condition. This suggests, on the one hand, that children were sensitive to lexical frequency but not to the argument vs. adjunct structure manipulation introduced in the study, and, on the other hand, that, at this developmental stage, children may still struggle with clitic processing depending on their level of grammatical competence. Concerning our experimental manipulation, these findings replicate the results obtained with 11-year-old children in Experiment 1, where no effect of condition was found either. Together, the two experiments suggest that, at this developmental stage, the same effort is required for children to process verb constructions containing arguments and adjuncts realised through clitics, contrary to what happens for adolescents and adults. The absence of an effect of our experimental manipulation in the eye-tracking paradigm indicates that the result obtained in Experiment 1 is unlikely to be solely attributable to methodological limitations of the self-paced reading paradigm.

5. General Discussion

Across two experiments, we investigated the processing of verb constructions that contain a dative clitic serving as either an argument or an adjunct. Their processing was tested with children, adolescents, and adults. In Experiment 1, which employed a self-paced reading paradigm, it was shown that, in a sentence containing a pre-verbal dative clitic, trivalent verbs selecting a dative argument are read faster than bivalent verbs allowing for the presence of a benefactive dative by both adults and fourteen-year-old adolescents. This finding is consistent with the Argument Structure Hypothesis made by Boland and Blodgett (2006), which predicts that verb constructions with arguments should be processed faster than those with adjuncts, since arguments can be lexically retrieved, whereas adjuncts require additional syntactic computation. While Boland and Blodgett’s original study involved an assessment of adults’ processing of arguments and adjuncts presented as prepositional phrases, the present study can expand their account in two important ways:

(i) by showing that such account is also compatible with adults' processing of sentences containing dative clitics as arguments or adjuncts; and (ii) by providing evidence for a protracted developmental trajectory in the processing of these constructions. In fact, while fourteen-year-old adolescents behaved in line with adult participants during self-paced reading, eleven-year-old children did not process the two constructions differently. Experiment 2 replicated this finding with ten-year-old children using an eye-tracking-while-reading paradigm, showing that they displayed no advantage in processing argument structures compared to structures presenting dative clitics as adjuncts.

Taken together, the two studies converge in showing that, unlike adolescents and adults, children at the ages of ten and eleven process structures presenting argument and adjunct dative clitics similarly. Given that it was replicated with different methodologies, this finding is unlikely to be explained by limitations of the experimental paradigms used. Instead, it suggests that sensitivity to the distinction between arguments and adjuncts in real-time processing still develops during later childhood, at least when these are presented in the form of pre-verbal dative clitics, and that their processing becomes adult-like only during the first years of adolescence. A possible explanation for this developmental pattern lies in children's grammatical competence. The results of our eye-tracking paradigm show that it is the grammatical competence rather than differences in the verb's argument structure that determine children's behaviour, as lower levels of grammatical ability were associated with more regressions on dative clitics (as discussed in Figure 2), reflecting the increased processing effort to integrate the information they encode. The Argument Structure Hypothesis (Boland & Blodgett, 2006), which our data on adults support, posits that syntactic derivation should play a role in processing only in the case of adjuncts, and this is what causes the increased costs tied to this condition. It could be hypothesised that, at least in the presence of a clitic, pre-adolescent children process the verb structure through syntactic derivation regardless of the distinction between argument and adjunct. It is worth recalling that this distinction is also reflected in the syntactic differences between high and low applicative phrases, as discussed in the introduction, where the syntactic structure plays a central role alongside the lexical one. Given how previous research shows that children aged 3–4 and 5–6 can already associate verbs with their argument structure (Peter et al., 2015), the similar treatment of argument and adjunct dative clitics detected here at the ages of ten and eleven should not, in principle, be due to difficulties with the argument structure per se, but may rather reflect still-maturing grammatical representations, especially while processing clitics, whose development has been shown to continue across childhood (Moscati et al., 2023).

As an anonymous reviewer suggests, some other factors could, in principle, play a role in explaining our results, for example, some hypotheses on the role of the syntactic space dedicated to the interpretation of discourse information, i.e., the left periphery (Rizzi, 1997), such as the Growing Trees hypothesis (Friedmann et al., 2021) and the Interface Hypothesis (Sorace & Serratrice, 2009). However, both approaches seem to not be adequate in several respects. More specifically, the Growing Trees hypothesis is typically used to explain the grammar of very young children (usually at the beginning of primary-school period, as in the original paper by (Friedmann et al., 2021)) who are still building or stabilising functional categories. In our study, we tested older children (the youngest children were ten-year-olds) and adolescents. At this developmental stage, the whole syntactic tree—including the CP layer and the left periphery—is already fully mature and grammatically adult-like. Therefore, the null effect observed in our ten- and eleven-year-old groups cannot be attributed to a lack of structural or grammatical knowledge regarding clitic structures or clitic movement. On the other hand, the Interface Hypothesis (Sorace & Serratrice, 2009) is an elegant framework for explaining why external interfaces (such as syntax-

discourse/pragmatics) show vulnerability or prolonged optionality compared to internal ones (syntax-semantics). Crucially, this hypothesis was mainly formulated to address language attrition, L2 acquisition, or bilingual development. Since our participants are all native, monolingual Italian speakers, and the processing of argument vs. adjunct clitics relies heavily on internal configurations (lexical specification vs. high/low applicative syntactic derivation), an external syntax-discourse mismatch is unlikely to explain the processing differences between late childhood and early adolescence. Therefore, these approaches do not seem to be adequate either for our participants or for the phenomenon we tested.

Moreover, our findings cannot be explained by frequency-based accounts either. More specifically, we found an effect of the condition in adults and adolescents, but not in ten- to eleven-year-old children, a population where the role of frequency in the linguistic system is known to be particularly relevant. In fact, we found a crucial pattern: (i) only children showed an effect of verb frequency, and (ii) only children did not show differences in processing argument vs. adjunct clitics. Thus, children are particularly sensitive to frequency, more than adolescents and adults, while they are not to the argument vs. adjunct structure manipulation introduced in the study. Therefore, frequency alone is not a good predictor for the difference between arguments and adjuncts.

It must also be mentioned that, overall, reading tasks are more effortful for children than they are for adults or adolescents. A progressive difficulty in performing the self-paced reading task can be observed by looking at the overall differences in reading times among the three groups tested (as depicted in Figure 1): children are generally slower than adolescents, who are, in turn, slower than adults. This is, in itself, not unexpected; however, it could provide a complementary explanation for the developmental pattern observed in the data. The effort required to perform the reading task specifically could have made it more difficult for children, but not for more proficient readers, to put in place efficient processing strategies, such as lexical retrieval, leading them to resort to more uniform, computation-based strategies for both arguments and adjuncts. Such an interpretation does not contrast with the one provided above about grammatical competence, as the difficulty of the task could be enhanced also by syntactic complexity, for instance, when movement operations, such as cliticisation, are involved. This would be true especially if, as hypothesised, the relevant grammatical representations are still maturing at the age tested. These findings highlight the importance of considering grammatical knowledge and the relative demands of a task, especially if it involves reading, when evaluating not only children's syntactic development, but also their language processing skills more generally. This is particularly relevant in the school setting, where much information is provided through written language, and, thus, processing and comprehension must be coordinated simultaneously during reading.

A limitation of the present study concerns the sample sizes of the child groups. Although these sample sizes are modest, it should be observed that the number of observations per condition is comparable to that of previous psycholinguistic studies with children of similar ages (e.g., Bentea & Marinis, 2021; Joseph & Liversedge, 2013; Wonnacott et al., 2016). Nevertheless, the smaller sample sizes in the child groups relative to the adolescent and adult groups warrant some caution in interpreting the null effects observed in Experiment 1 and the pattern of the results in Experiment 2. Future work should seek to replicate these findings with larger child samples and, ideally, include a formal a priori power analysis informed by effect sizes reported in comparable developmental reading studies.

Taken together, our results point to the importance of integrating insights from formal linguistic theory and psycholinguistic research to better understand how syntactic structures develop in first language acquisition, and how they are processed during reading. In

this perspective, the present study shows that the distinction between arguments and adjuncts, when these are realised as dative clitics, is reflected in real-time sentence processing only once a sufficient level of syntactic and reading proficiency has been reached, and, in doing so, highlights a protracted developmental trajectory.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/languages11070146/s1>, The experimental stimuli for both experiments, as well as the datasets containing the results and the full analysis scripts, can be found in our online repository on OSF (<https://osf.io/8mvua>, accessed on 24 June 2026).

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Informed Consent Statement: The study was approved by the Ethics Committee of the University of Pavia and informed consent was obtained from adult participants and the parents or legal guardians of the minors; assent was also obtained from adolescent and child participants, and it was made clear to them that they could withdraw from participating in the study at any moment, without consequence.

Data Availability Statement: The experimental stimuli for both experiments, as well as the datasets containing the results and the full analysis scripts, can be found in our online repository on OSF (<https://osf.io/8mvua>, accessed on 24 June 2026).

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Notes

- ¹ In this work, we will not consider monovalent verbs in co-occurrence with dative clitics because, in this case, the non-argumental clitic can more easily fulfill other functions, such as the Ethical Dative. Consider, for example, a sentence like “Gigi *mi* è arrivato terzo” (literally, ‘Louis CL is arrived third’, meaning ‘Louis has arrived in third position’), where the monovalent verb *arrivare* (‘to arrive’) hosts the clitic *mi* that looks like a benefactive but is not, as it is an instance of an ethical dative. This is witnessed by the fact that it cannot be replaced by means of the corresponding prepositional phrase *a me* (‘to me’): “Gigi è arrivato terzo *a me*” (literally, ‘Louis is arrived third to me’), contrary to what benefactives usually allow (see sentence 4). See Greco (2025) and the references therein for a distinction between the Ethical Dative and the Benefactive.
- ² As an anonymous reviewer pointed out, an alternative perspective can in principle be offered by some frequency approaches, where processing advantages are guided by distributional patterns in the language input. See the final section for a more detailed discussion.

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